25 The straight line **L** has equation $3x + 2y = 17$

The point A has coordinates (0, 2)
The straight line **M** is perpendicular to **L** and passes through A.

Line **L** crosses the y-axis at the point B.
Lines **L** and **M** intersect at the point C.

Work out the area of triangle ABC.
You must show all your working.

L: 3x + 2y = 17 gives y = −(3/2)x + 8.5.

M is perpendicular: gradient 2/3; through A(0,2),

so M: y = (2/3)x + 2.

At C: −(3/2)x + 8.5 = (2/3)x + 2.

Thus x = 3, y = 4, so C = (3,4).

B = (0,8.5), giving AB = 8.5 − 2 = 6.5.

Area ABC = ½ × 6.5 × 3 = 9.75 units^2.

5 marks total
Gradient L = −3/2
M gradient 2/3
B has y = 8.5
Find C = (3,4)
Area 9.75

(Total for Question 25 is 5 marks)